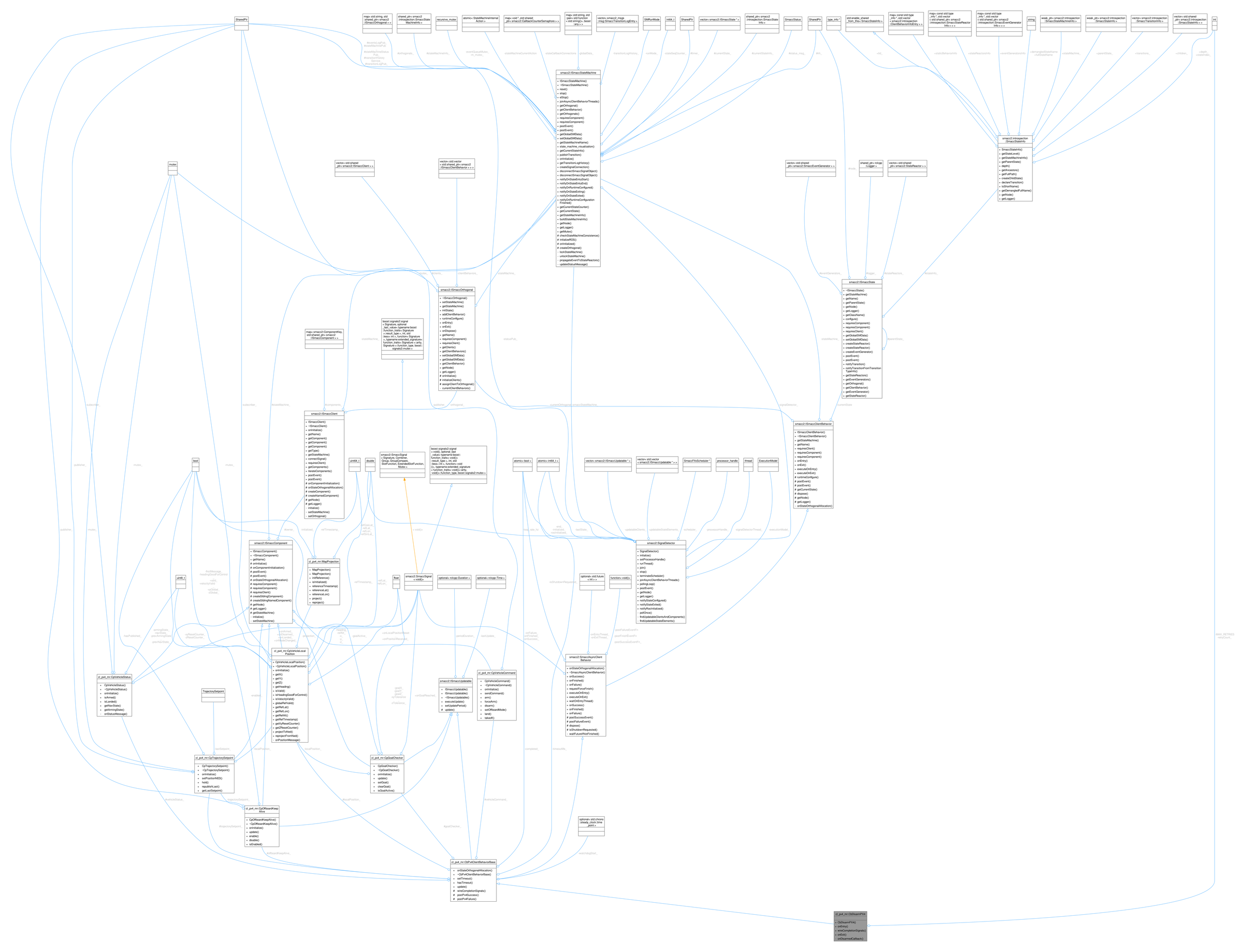
SharedPtr
map< std::string, std ::shared_ptr< smacc2 ::ISmaccOrthogonal > >
shared_ptr< smacc2 ::introspection::SmaccState MachineInfo >
recursive_mutex
atomic< StateMachineInternal Action >
map< void *, std::shared _ptr< smacc2::CallbackCounterSemaphore > >
map< std::string, std ::shared_ptr< action < std::string(), boost ::any > >
vector< smacc2::msgs ::msg::SmaccTransitionLogEntry >
SMRunMode
int64_t
SharedPtr
vector< smacc2::ISmaccState * >
shared_ptr< smacc2 ::introspection::SmaccState Info >
SmaccStatus
SharedPtr
type_info *
std::enable_shared _from_this< SmaccStateInfo >
map< const std::type _info *, std::vector < smacc2::introspection ::ClientBehaviorInfoEntry > >
map< const std::type _info *, std::vector < std::shared_ptr< smacc2 ::introspection::SmaccStateReactor Info > > >
map< const std::type _info *, std::vector < std::shared_ptr< smacc2 ::introspection::SmaccEventGenerator Info > > >
string
weak_ptr< smacc2::introspection ::SmaccStateMachineInfo >
weak_ptr< smacc2::introspection ::SmaccStateInfo >
vector< smacc2::introspection ::SmaccTransitionInfo >
vector< std::shared _ptr< smacc2::introspection ::SmaccStateInfo > >
int
#eventsLogPub_
#stateMachineStatus Pub_
#transitionHistory _service_
#transitionLogPub_
#orthogonals_
#stateMachineInfo_
-eventQueueMutex_ m_mutex_
+stateMachineCurrentAction
+stateCallbackConnections
-globalData_
+transitionLogHistory_
+runMode_
+stateSeqCounter_
#timer_
#currentState_
#currentStateInfo_
#status_msg_
#sh_
+list_
+stateBehaviorInfo
+stateReactorsInfo
+eventGeneratorsInfo
+demangledStateName +fullStateName
+stateMachine_
+parentState_
+transitions_
+children_
+depth_ +stateIndex_
smacc2::introspection ::SmaccStateInfo
+ SmaccStateInfo()
+ getStateLevel()
+ getStateMachineInfo()
+ getParentState()
+ depth()
+ getAncestors()
+ getFullPath()
+ createChildState()
+ declareTransition()
+ toShortName()
+ getDemangledFullName()
+ getNode()
+ getLogger()
smacc2::ISmaccStateMachine
+ ISmaccStateMachine()
+ ~ISmaccStateMachine()
+ reset()
+ stop()
+ eStop()
+ joinAsyncClientBehaviorThreads()
+ getOrthogonal()
+ getClientBehavior()
+ getOrthogonals()
+ requiresComponent()
+ requiresComponent()
+ postEvent()
+ postEvent()
+ getGlobalSMData()
+ setGlobalSMData()
+ getStateMachineName()
+ state_machine_visualization()
+ getCurrentStateInfo()
+ publishTransition()
+ onInitialize()
+ getTransitionLogHistory()
+ createSignalConnection()
+ disconnectSmaccSignalObject()
+ disconnectSmaccSignalObject()
+ notifyOnStateEntryStart()
+ notifyOnStateEntryEnd()
+ notifyOnRuntimeConfigured()
+ notifyOnStateExiting()
+ notifyOnStateExited()
+ notifyOnRuntimeConfiguration Finished()
+ getCurrentStateCounter()
+ getCurrentState()
+ getStateMachineInfo()
+ buildStateMachineInfo()
+ getNode()
+ getLogger()
+ getMutex()
# checkStateMachineConsistence()
# initializeROS()
# onInitialized()
# createOrthogonal()
- lockStateMachine()
- unlockStateMachine()
- propagateEventToStateReactors()
- updateStatusMessage()
mutex
vector< std::shared _ptr< smacc2::ISmaccClient > >
vector< std::vector < std::shared_ptr< smacc2 ::ISmaccClientBehavior > > >
vector< std::shared _ptr< smacc2::SmaccEventGenerator > >
shared_ptr< rclcpp ::Logger >
vector< std::shared _ptr< smacc2::StateReactor > >
#mutex_
#clients_
+clientBehaviors_
+stateMachine_
#eventGenerators_
#logger_
#stateReactors_
#stateInfo_
smacc2::ISmaccOrthogonal
+ ~ISmaccOrthogonal()
+ setStateMachine()
+ getStateMachine()
+ initState()
+ addClientBehavior()
+ runtimeConfigure()
+ onEntry()
+ onExit()
+ onDispose()
+ getName()
+ requiresComponent()
+ requiresClient()
+ getClients()
+ getClientBehaviors()
+ setGlobalSMData()
+ getGlobalSMData()
+ getClientBehavior()
+ getNode()
+ getLogger()
# onInitialize()
# initializeClients()
# assignClientToOrthogonal()
- currentClientBehaviors()
smacc2::ISmaccState
+ ~ISmaccState()
+ getStateMachine()
+ getName()
+ getParentState()
+ getNode()
+ getLogger()
+ getClassName()
+ configure()
+ requiresComponent()
+ requiresComponent()
+ requiresClient()
+ getGlobalSMData()
+ setGlobalSMData()
+ createStateReactor()
+ createStateReactor()
+ createEventGenerator()
+ postEvent()
+ postEvent()
+ notifyTransition()
+ notifyTransitionFromTransition TypeInfo()
+ getStateReactors()
+ getEventGenerators()
+ getOrthogonal()
+ getClientBehavior()
+ getEventGenerator()
+ getStateReactor()
map< smacc2::ComponentKey, std::shared_ptr< smacc2 ::ISmaccComponent* > >
boost::signals2::signal < Signature, optional _last_value< typename boost ::function_traits< Signature >::result_type >, int, std ::less< int >, function< Signature >, typename extended_signature< function_traits< Signature >::arity_ , Signature >::function_type, boost:: signals2::mutex >, ... >
#stateMachine_
#statusPub_
#eventGenerators_
#logger_
#stateReactors_
#currentState_
smacc2::ISmaccClientBehavior
+ ISmaccClientBehavior()
+ ~ISmaccClientBehavior()
+ getStateMachine()
+ getName()
+ requiresClient()
+ requiresComponent()
+ requiresComponent()
+ onEntry()
+ onExit()
+ executeOnEntry()
+ executeOnExit()
# runtimeConfigure()
# postEvent()
# postEvent()
# getCurrentState()
# dispose()
# getNode()
# getLogger()
- onStateOrthogonalAllocation()
#subscriber_
#stateMachine_
#components_
#publisher_
#orthogonal_
-currentOrthogonal_
-smaccStateMachine_
#signalDetector_
#currentState_
smacc2::ISmaccClient
+ ISmaccClient()
+ ~ISmaccClient()
+ onInitialize()
+ getName()
+ getComponent()
+ getComponent()
+ getComponent()
+ getType()
+ getStateMachine()
+ connectSignal()
+ requiresClient()
+ getComponents()
+ iterateComponents()
+ postEvent()
+ postEvent()
# onComponentInitialization()
# onStateOrthogonalAllocation()
# createComponent()
# createNamedComponent()
# getNode()
# getLogger()
- initialize()
- setStateMachine()
- setOrthogonal()
bool
uint64_t
double
smacc2::SmaccSignal < Signature, Combiner, Group, GroupCompare, SlotFunction, ExtendedSlotFunction, Mutex >
boost::signals2::signal < void(), optional_last _value< typename boost:: function_traits< void()>::result_type >, int, std ::less< int >, function< void ()>, typename extended_signature < function_traits< void()>::arity_ , void()>::function_type, boost::signals2::mutex >, ... >
atomic< bool >
atomic< int64_t >
vector< smacc2::ISmaccUpdatable* >
vector< std::vector < smacc2::ISmaccUpdatable* > >
SmaccFifoScheduler*
processor_handle
thread
ExecutionModel
#publisher_
#mutex_
#subscriber_
#stateMachine_
#components_
#publisher_
#orthogonal_
-currentOrthogonal_
-smaccStateMachine
#signalDetector_
#currentState_
smacc2::ISmaccComponent
+ ISmaccComponent()
+ ~ISmaccComponent()
+ getName()
# onInitialize()
# onComponentInitialization()
# postEvent()
# postEvent()
# onStateOrthogonalAllocation()
# requiresComponent()
# requiresComponent()
# requiresClient()
# createSiblingComponent()
# createSiblingNamedComponent()
# getNode()
# getLogger()
# getStateMachine()
- initialize()
- setStateMachine()
cl_p4_mr::MapProjection
+ MapProjection()
+ MapProjection()
+ initReference()
+ isInitialized()
+ referenceTimestamp()
+ referenceLat()
+ referenceLon()
+ project()
+ reproject()
uint8_t
float
smacc2::SmaccSignal < void()>
optional< rclcpp::Duration >
optional< rclcpp::Time >
optional< std::future < int > >
function< void()>
smacc2::SignalDetector
+ SignalDetector()
+ initialize()
+ setProcessorHandle()
+ runThread()
+ join()
+ stop()
+ terminateScheduler()
+ joinAsyncClientBehaviorThreads()
+ pollingLoop()
+ postEvent()
+ getNode()
+ getLogger()
+ notifyStateConfigured()
+ notifyStateExited()
+ notifyIfAllInitialized()
+ pollOnce()
- findUpdatableClientsAndComponents()
- findUpdatableStateSemantics()
#owner_
#initialized_
-refTimestamp_
-updatableClients_
-updatableStateSemantics_
-scheduler_
-processorHandle_
-signalDetectorThread_
-executionModel_
-refLat_ -refLon_ -refAlt_ -refUnk_
-refTimestamp_
-isShutdownRequested_
#end_ -initialized_ -wasInitialized_
-lastState_
#node_rate_hz
#MAX_RETRIES retryCount_
#firstMessage_ -headingGoodForControl
-valid -velocityValid_
-xyGlobal_ -zGlobal_
-armingState_ -navState_ -prevArmingState_
-prevNavState_
-hasPublished_
-xyResetCounter_ -zResetCounter_
-armed_ -xyArmed_ -zArmed_ -wasLanded_ -xyMultipleChanged_
-zMultipleChanged_
-timeout_
-reading_ -refAlt_ -x_ -y_ -z_
-goalActive_
+onLocalPositionReset
+onPositionReported_
-periodDuration_
-lastUpdate_
+onFailure_ -onFinished_ -onSuccess_
-postFailureEventFn
-postFinishEventFn
-postSuccessEventFn_
-onEntryThread_ -onExitThread_
cl_p4_mr::CpVehicleStatus
+ CpVehicleStatus()
+ ~CpVehicleStatus()
+ onInitialize()
+ isArmed()
+ isLanded()
+ getNavState()
+ getArmingState()
- onStatusMessage()
cl_p4_mr::CpVehicleLocal Position
+ CpVehicleLocalPosition()
+ ~CpVehicleLocalPosition()
+ onInitialize()
+ getX()
+ getY()
+ getZ()
+ getHeading()
+ isValid()
+ isHeadingGoodForControl()
+ isVelocityValid()
+ globalRefValid()
+ getRefLat()
+ getRefLon()
+ getRefAlt()
+ getRefTimestamp()
+ getXyResetCounter()
+ getZResetCounter()
+ projectToNed()
+ reprojectFromNed()
- onPositionMessage()
smacc2::ISmaccUpdatable
+ ISmaccUpdatable()
+ ISmaccUpdatable()
+ ~ISmaccUpdatable()
+ execute(Update)
+ setUpdatePeriod()
# update()
cl_p4_mr::CpVehicleCommand
+ CpVehicleCommand()
+ ~CpVehicleCommand()
+ onInitialize()
+ sendCommand()
+ arm()
+ forceArm()
+ disarm()
+ setOffboardMode()
+ land()
+ takeoff()
smacc2::SmaccAsyncClient Behavior
+ onStateOrthogonalAllocation()
+ ~SmaccAsyncClientBehavior()
+ onSuccess()
+ onFinished()
+ onFailure()
+ requestForceFinish()
+ executeOnEntry()
+ executeOnExit()
+ waitOnEntryThread()
+ onSuccess()
+ onFinished()
+ onFailure()
# postSuccessEvent()
# postFailureEvent()
# dispose()
# isShutdownRequested()
- waitFutureIfNotFinished()
TrajectorySetpoint
-enabled_
-lastSetpoint_
-localPosition_
-localPosition_
-localPosition_
-goalX_ -goalY_ -goalZ_ -xyTolerance_ -zTolerance_
+onGoalReached_
-completed_
-timeoutMs_
cl_p4_mr::CpTrajectorySetpoint
+ CpTrajectorySetpoint()
+ ~CpTrajectorySetpoint()
+ onInitialize()
+ setPositionNED()
+ hold()
+ republishLast()
+ getLastSetpoint()
cl_p4_mr::CpGoalChecker
+ CpGoalChecker()
+ ~CpGoalChecker()
+ onInitialize()
+ update()
+ setGoal()
+ clearGoal()
+ isGoalActive()
#vehicleStatus_
#trajectorySetpoint_
#localPosition_
#vehicleCommand_
#trajectorySetpoint_
#goalChecker_
cl_p4_mr::CpOffboardKeep Alive
+ CpOffboardKeepAlive()
+ ~CpOffboardKeepAlive()
+ onInitialize()
+ update()
+ enable()
+ disable()
+ isEnabled()
optional< std::chrono ::steady_clock::time _point >
#offboardKeepAlive_
#watchdogStart_
cl_p4_mr::CpPx4ClientBehaviorBase
+ onStateOrthogonalAllocation()
+ ~CpPx4ClientBehaviorBase()
+ setTimeout()
+ hasTimeout()
+ update()
# wireCompletionSignals()
# postPx4Success()
# postPx4Failure()
cl_p4_mr::CbDisarmPX4
+ CbDisarmPX4()
+ onEntry()
+ wireCompletionSignals()
+ onExit()
- onDisarmedCallback()